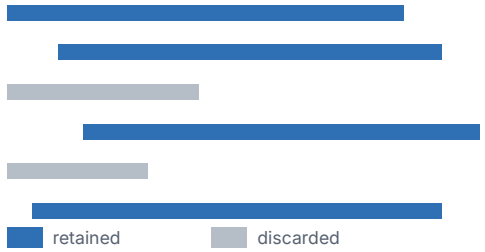
1 Input and filtering

FASTQ or FASTQ.GZ files, or a MinKNOW fastq_pass folder (one sample per barcode)

Read length ≥ 100 bp (user-defined)

Optional minimum mean read quality (Q)



2 Orientation

Each read is oriented on the HTT sense strand: both strands are analysed

Forward-strand reads (CAG)n

Reverse-strand reads (CTG)n

reverse-complement → sense orientation

Reverse-strand reads: ≈38% of parsed reads

3 Anchoring

Each read is anchored on three sequences



1. 5′ flank CTCAAGTCCTTC (1 mismatch tolerated)
2. start of the CCG repeat
3. CCT anchor

Anchors 1–2 delimit the CAG tract and the adjacent sequence (adj.); CAG count = tract / 3

4 Configuration

Adjacent sequence: exact match after ≥2 CAG codons

canonical	CAACAGCCGCCA
CAA-loss	CCGCCA
CCA-loss	CAACAG
CAACAGCCGCCA-loss	none
CAACAG-dup (G2)	CAACAGCAACAGCCGCCA
CAA-gain G1	CAACAACAGCCGCCA
CAA-gain G3–G8	CAA(CAG)k-1CAACAGCCGCCA
unclassified	other sequence

Reads with an unresolved junction are not used

reads with a resolved structure: uninterrupted CAG count, configuration and CCG count

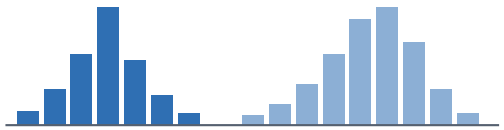
5 Allele discovery

Modes of the CAG distribution, searched separately within each configuration

Score: reads within ±1 CAG (±2 above 35)

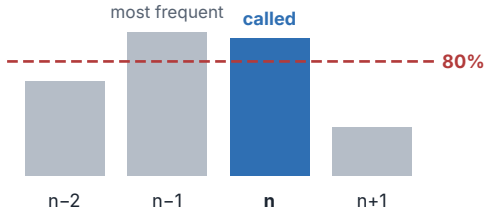
Called if ≥5% of resolved reads and ≥10 reads

At most two alleles



6 Stutter-aware sizing (≥36 CAG)

PCR stutter makes n−1 almost as abundant as n: the allele is the longest count with ≥80% of the reads of the most frequent count



7 Phasing and fragment equivalent

CCG length and configuration of each allele = majority of its reads; chimeric reads ignored

Allele 1	17 CAG	canonical	CCG 7
Allele 2	39 CAG	canonical	CCG 7

Fragment-equivalent CAG = codons from the 5′ anchor to the CCG-CCA motif − 4

8 Output and flags

summary.tsv · result.json · haplotype counts

HTML report with CAG, CAG×CCG and read plots

- Call differs from most frequent count (±1)
- Alleles one repeat apart not separated
- Marked allelic imbalance (ratio < 0.33)
- Unclassified sequence: confirm independently